Physikalisches Grundpraktikum

Versuchsprotokoll: Stehende Schallwellen in Luft

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Gruppe C4

Stehende Schallwellen in Luft

Aufgabe 1 Resonanzfrequenzen mit Cassy

Geben Sie die Rohrlänge L und ihre Unsicherheit an. Stellen Sie Ihre Daten als U-f-Diagramm dar: Zeichnen Sie Ihre Messpunkte ein und verbinden Sie diese ggfs. mit Linien. Lesen Sie die Resonanzfrequenzen f_n ab. Bestimmen Sie die Messunsicherheit auf die f_n (Wiederholungsmessung, ersatzweise Abschätzung der Genauigkeit, mit der die Peakpositionen abgelesen werden können). Zeigen Sie für jede Resonanzfrequenz einen vergrößerten Ausschnitt Ihres Diagramms, in den Sie die gefundene Peakposition und ihre Unsicherheit einzeichnen. Tragen Sie die f_n gegen n auf und führen Sie eine lineare Regression durch. Bestimmen Sie die Schallgeschwindigkeit und ihre Unsicherheit.

First we measured the length of the tube we used in our experiment, we obtained the following results conducting independent experiments:

Length	Measurements		Mean
L	$41{,}95$ cm	42 cm	$41{,}975$ cm

we then calculate the uncertainty in our length measurement by quadratically adding our stadard deviation and the sys. error given by the manufacturer of the tool of measurement (in this case a measuring tape with 0,7 mm uncertainty):

$$\sigma_L = \sqrt{\sigma^2 + \sigma_{sys}^2}$$

$$\sigma_L = \sqrt{0{,}35^2 + (\frac{0{,}07}{\sqrt{3}})^2} \text{ mm} = 0{,}357 \text{ mm}$$

we then used scripts to find the first five resonance frequencies, we then sorted our data and noticed that in some of our Measurements we had more than one voltage corresponding to a frequency value so we averaged them and took the

peaks of the resulting plot for clarity. for each of our peaks, we decided that an uncertainty of 2 Hz is enough to contain the top $10\% - 20\%$ of our peak length, and so we took 2 Hz as conservative frequency uncertainty (highlighted with a red error bar). We then plot our found frequencies and their uncertainties in dedicated plots:

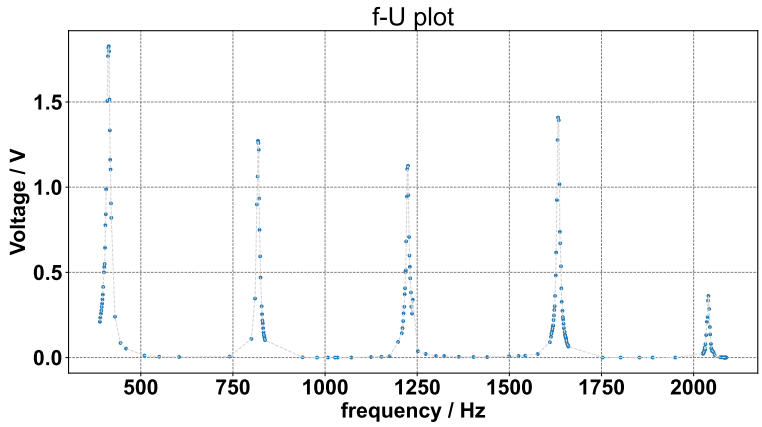

Abbildung 1: plot for frequency and voltage

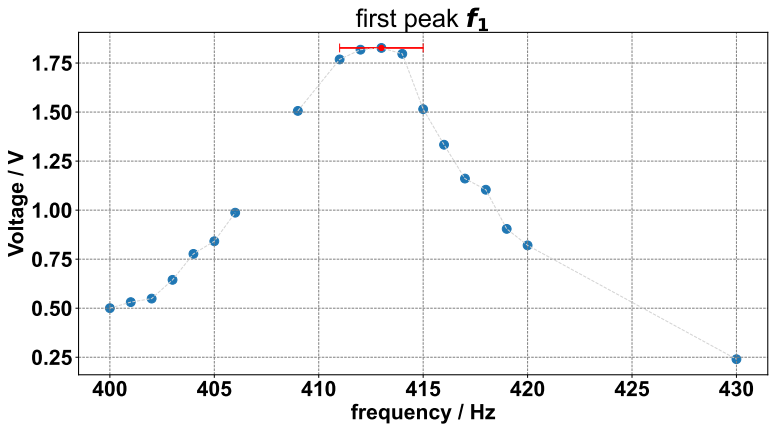

Abbildung 2: first frequency

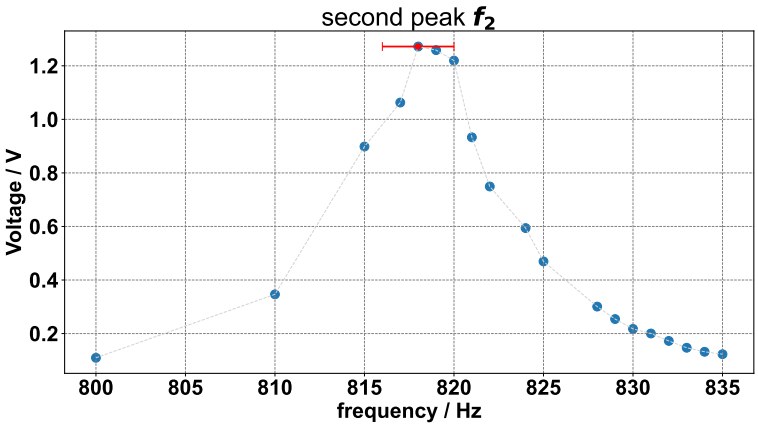

Abbildung 3: second frequency

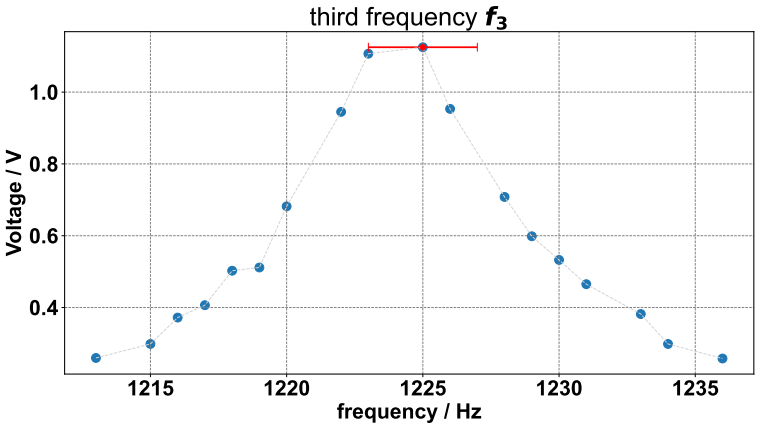

Abbildung 4: third frequency

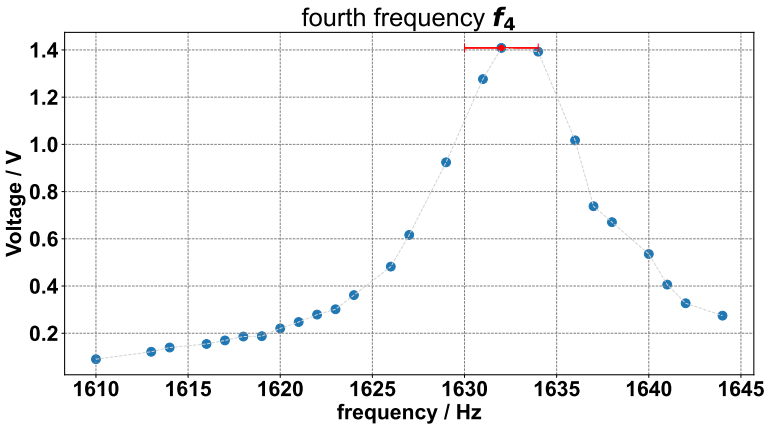

Abbildung 5: fourth frequency

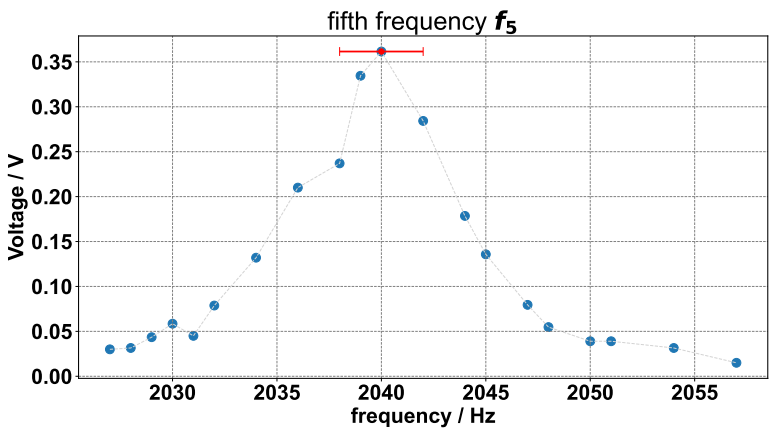

Abbildung 6: fifth frequency

we then give the results of our maximum peaks as follows:

- 413 Hz: first resonance frequency.
- 818 Hz: second resonance frequency.
- 1225 Hz: third resonance frequency.
- 1632 Hz: fourth resonance frequency.
- 2040 Hz: fifth resonance frequency.

Then through a linear regression we fit our resonance frequencies to their number, we obtain the following result:

```
R = (406.8+/-0.6) Hz,
b = (5.2+/-2.1) Hz
chi2/dof = 0.7 / 3
corr = -0.904535
```

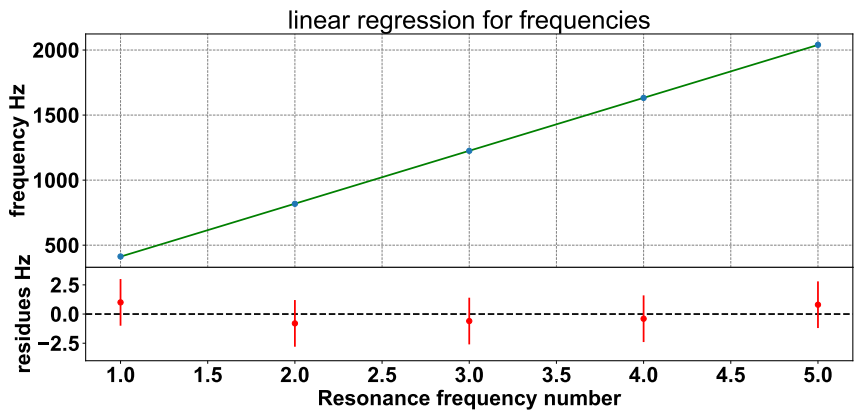

Abbildung 7: linear regression fit for frequencies and their numbers

We then combined our frequencies divided by their number in a variable f_n/n and calculated its uncertainty through weighted average of the resulting division, overall we obtained the following variables and their uncertainties, which leads us to the speed of sound:

$$v = \frac{f_n}{n} \cdot 2L$$

```
L: 41,975 +/- 0,0357 cm
f_n / n : 408,2 +/- 0,27 Hz
v : 342,70 +/- 0,37 m/s
```

Aufgabe 2 Resonanzfrequenzen mit dem Oszilloskop

Tabellieren Sie die gefundenen Resonanzfrequenzen f_n. Zur Bestimmung der Messunsicherheit auf die f_n führen Sie Wiederholungsmessungen durch oder schätzen Sie die Genauigkeit ab, mit der Sie die Resonanzfrequenzen mit dem Oszilloskop finden können. Bestimmen Sie aus Ihren Messdaten die Schallgeschwindigkeit und ihre Unsicherheit. Tragen Sie dazu die f_n gegen n auf und führen Sie eine lineare Regression durch.

We triggered on CH1 which we connected the Function Generator to with the settings Typ= Flanke, Flanke=positiv, Modus=normal, Kopplung=HF Reject. The microphone was connected to CH2. We used the cursor Typ Amplitude to find the maximum of the microphone signal and found the following resonance frequencies:

n	1	2	3	4	5
f_n /Hz	412	818	1223	1632	2039
f_n/n /Hz	412	409	407,6	408	407,8

We noticed finding the true maximum is not easy to distinguish, which is why we are estimating the true maximum to be within one Hertz. So our statistical error is $\sigma_{stat} = 1Hz$. The systematic error due to the oscilloscope frequency response is estimated at up to 30% of the measured amplitude near the -3 dB point but since our measurements are well below the bandwidth this error is negligible.
To calculate the speed of sound we use the following formula again

$$v_n = \frac{2Lf_n}{n}.$$

which gives us the following results:

```
L: 41,975 +/- 0,0357 cm
f_n / n : 408,00 +/- 0,13 Hz
v : 342,52 +/- 0,31 m/s
```

which gives us the mean $\overline{v} = 342{,}52$ m/s and the standard error of the mean $\sigma_{\overline{v},stat} = 0{,}31$ m/s. We performed a linear regression of the measured amplitude versus frequency.

```
R = (406.80+/-0.32) Hz
b = (4.4+/-1.0) Hz
chi2/dof = 4.4 / 3
corr = -0.904529
```

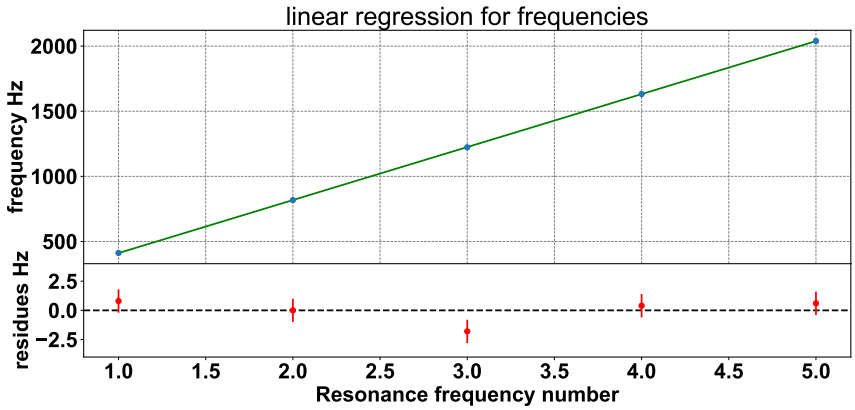

Abbildung 8: Linear Regression for the resonances with the oscillator

The slope R=406,8 Hz and intercept b=4.4 Hz are given by the fit. The dominant uncertainty comes from estimating the true maximum of the amplitude, which we account for separately as a statistical error.

Aufgabe 3 Vermessung der stehenden Welle

Tabellieren Sie die gefundenen Positionen x_n der Druckknoten und -bäuche. Geben Sie die jeweils verwendete Resonanzfrequenz f und ihre Unsicherheit an. Zur Bestimmung der Messunsicherheit auf die x_n führen Sie Wiederholungsmessungen durch oder schätzen Sie die Genauigkeit ab, mit der Sie die Druckknoten bzw. -bäuche durch Verschiebung des Mikrofons und Beobachtung der Amplitude am Oszilloskop einstellen können. Tragen Sie die Positionen x_n gegen n auf und führen Sie eine geeignete lineare Regression durch. Bestimmen Sie aus Ihren Ergebnissen die Schallgeschwindigkeit und ihre Unsicherheit. Berechnen Sie den gewichteten Mittelwert über die Ergebnisse aus Ihren Messreihen bei verschiedenen Resonanzfrequenzen.

We repeated the measurements with different settings of our apparatus in ordert to eliminate systemic errors, after a few tries, we obtained the following results:

```
f: 2043 Hz
```

Peaks	0 cm	8 cm	$16{,}9$ cm
Troughs	4 cm	$12{,}5$ cm	21 cm

for another resonance frequency, we have a less precise measurement, but we put it here to measure a weighted average:

```
f: 1228 Hz
```

Peaks	0 cm	15 cm
Troughs	7 cm	20,8 cm

We then analyse our data through a linear regression, we take a conservative error of 0,2 cm and 0.5 cm respectively, since we see noticable change in the amplitude after such change. lastly, we obtain the following results:

```
R = (4.29+/-0.06) cm
b = (-0.39+/-0.21) cm
chi2/dof = 1.7 / 3
corr = -0.904533
```

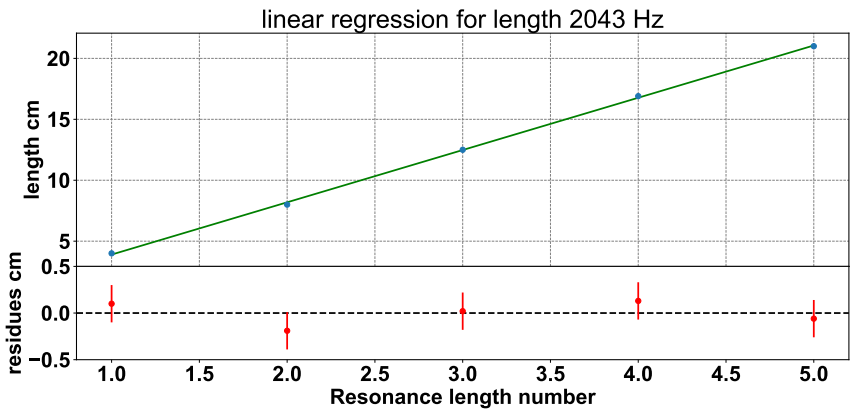

Abbildung 9: linear regression fit for lengths and their numbers 2043 Hz

```
R = (6.90+/-0.35) cm
b = (0.5+/-0.8) cm
 chi2/dof = 3.2 / 1
corr = -0.92582
```

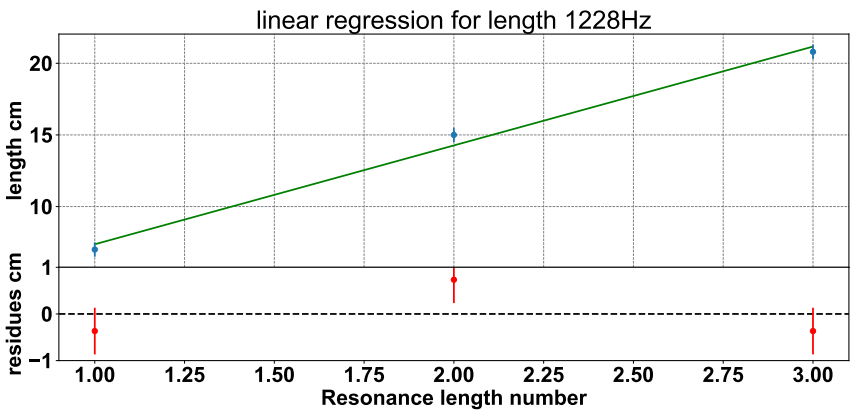

Abbildung 10: linear regression fit for lengths and their numbers 1228 Hz

we then simply calculate the resulting speed of sound thourgh the estimate of L_n/n through weighted averages and then inserting the results in the following equation:

$$v = \frac{2L_n}{n} \cdot 2f$$

speed	first result	second result	combined
v	341,58 $\pm$ 2,20 m/s	348,75 $\pm$ 6,53 m/s	342,29 $\pm$ 2,08 m/s

Aufgabe 4 Zusammenfassung und Vergleich

Stellen Sie die Ergebnisse aller drei Teilversuche nebeneinander. Berechnen Sie ggfs. ein gewichtetes Mittel der Schallgeschwindigkeit unter Berücksichtigung der Korrelationen der Unsicherheiten. Vergleichen Sie Ihr Ergebnis mit dem Literaturwert für die gemessene Raumtemperatur.

We first look at how our measurements can be correlated in their uncertainties:

```
result1: related to result2 by lenght measurement of the tube.
result2: related to result1 by length of tube.
result3: unrelated to both experiments.
```

So we need to calculated the weighted average of our results, while taking into consideration the correlation of our uncertainties:

result1	result2	result3	combined
342,70 $\pm$ 0,37 m/s	342,52 $\pm$ 0,31 m/s	342,29 $\pm$2,08 m/s	342,58 $\pm$ 0,31 m/s

The litrature value of the speed of sound at our temprature is around $345{,}47$ m/s. which is an offset by around 3 m/s. this hints at a systemic error in our measurements apparatus that has shifted our results by around 0.9%. This can be mainly due length or frequency uncertainties.